

(B) Layout-Protected EWC (L-EWC)

Trainable parameter groups



$$\mathbf{L} = \mathbf{L}_{\text{CE}} + \lambda_{\text{layout}} \mathbf{F}_{\text{layout}} (\theta - \theta^*)^2 + \lambda_{\text{other}} \mathbf{F}_{\text{other}} (\theta - \theta^*)^2$$

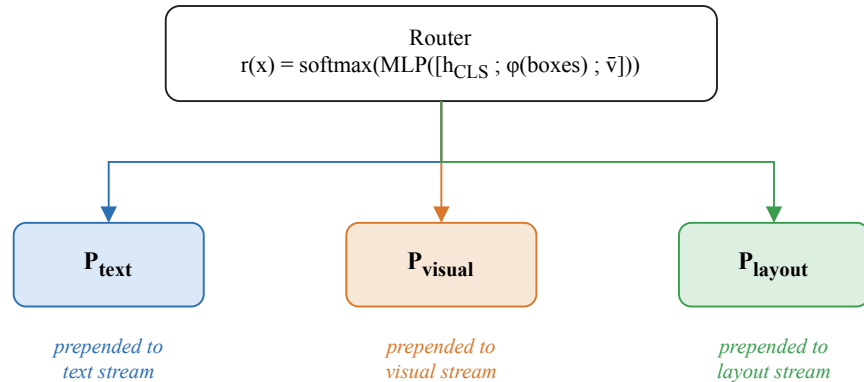
$$\lambda_{\text{layout}} \gg \lambda_{\text{other}}$$

Optional: layout-embedding distillation from previous-task model

Triggered if forgetting is uniform but 2-D position embeddings drift fastest; no inference-time parameters.

Legend: HIGH penalty (layout) Text sub-pool Visual sub-pool Layout sub-pool

(C) Modality-Routed Prompts



Triggered if per-scenario or per-modality forgetting patterns are dominant; sub-pools are modality-specific and independently trained.